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Many-body dissipative particle dynamics simulation of Newtonian and non-Newtonian nanodroplets spreading upon flat and textured substrates

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ABSTRACT

To deposit droplets on substrates efficiently is critical for many technological and industrial applications, which requires a systematic understanding of the droplet spreading process. In the paper, mesoscopic modeling of viscous nanodroplets spreading on different solid surfaces has been conducted by many-body dissipative particle dynamics. The influence of fluid viscosity has been particularly analyzed by changing the interaction parameters and the maximum droplet spreading diameter has been obtained. The dimensionless maximum spreading diameter is correlated to the Reynolds numbers (Re) and Weber numbers (We) as $\beta_{\max} = 0.7Re^{0.1445}We^{0.066}$. With the droplets partially attached by polymer chains to exhibit a shear-thinning non-Newtonian property, we demonstrate that the maximum spreading diameters are extremely different under a small initial kinetic energy, and gradually become consistent at a larger Weber number. Investigation of the spreading process on the intricate surfaces decorated by regular pillared structures shows a special expansion effect to improve the intrinsic wetting characteristic of the substrates, which is strongly controlled by the surface fraction of the pillared structures. Moreover, it is found that the fluid viscosity can profoundly affect the rebounding behavior of the droplet.

1. Introduction

Nanodroplet spreading on a solid surface is involved in a wide range of scientific and industrial applications including inkjet printing [1–3], nano-printing [4,5], precise spraying [6,7] and spray coating of photoresist [8]. The spreading process is a complex phenomenon with gas–liquid–solid interactions, and it remarkably affects the printing and spraying accuracies. Besides, the scale effect becomes significant when the droplet diameter changes from micrometer to nanometer. To control the droplet behaviors during the scientific and industrial applications, it is crucial to understand the spreading characteristics of nanodroplets on different solid surfaces.

Considerable efforts have been devoted to revealing the spreading characteristic of droplets both in numerical and experimental studies. Khojasteh et al. [9] reviewed that the droplet spreading characteristics are relevant to many factors including the physical properties of droplets (size, viscosity, density and surface tension), the initial kinetic energy of droplets and the substrate properties. Regarding the physical properties, the normal impact of water droplets on solid substrates was investigated experimentally by Son et al. [10]. The result showed that the final shape of a droplet is decided by the surface phenomenon. Mao et al. [11] studied the spread and rebound of droplets impacting on flat

surfaces experimentally, and found the maximum spreading diameter of a droplet is dependent on the liquid viscosity and impact velocity. As to the initial kinetic energy, Eggers et al. [12] studied the impact of a droplet onto a planar solid surface at high speed, and compared the theory of droplets spreading with numerical simulations. Clanet et al. [13] found a maximal spreading scales as an exponential relationship with Weber number related to the shock. For the substrate properties, there is evidence that the roughness of substrates plays a crucial role in regulating the droplet spreading [14,15]. Li and Zhang [16] investigated dynamic behavior of water droplets on the prepared superhydrophobic surface with dandelion-like structures, and analyzed the reason why water droplets can complete rebound from the superhydrophobic surface after impacting. Liu et al. [17] showed how superhydrophobic surfaces decorated with nano-textures can generate a counter-intuitive bouncing regime. Wang and Chen [18] explored the mechanism of droplets impinging on textured surfaces numerically, and obtained the numerical results consistent with the theoretical prediction. Kim et al. [19] reported the droplet impact characteristics on specific surfaces with different structures, and showed that microstructure can greatly affect the droplet spreading and retracting processes.

The spreading process of a Newtonian droplet has been thoroughly

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investigated from experimental observations and numerical simulations. However, understanding the dynamics of a non-Newtonian droplet is crucial for controlling the droplet spreading. Son and Kim [20] investigated the characteristic of the non-Newtonian droplets experimentally, and found it to be qualitatively different from those of Newtonian droplets. Bartolo et al. [21] examined the retraction dynamics of aqueous liquid droplets. They showed polymer additives can totally prevent the bouncing of liquid droplets, and demonstrated that non-Newtonian effects can extremely change the contact line. Despite it being explored by several experiments, few works of literature computationally conduct the spreading mechanisms of a non-Newtonian droplet. Moreover, when the droplet diameter decays to sub-micrometer or even nanometers, the spreading mechanisms can not be implemented by the traditional numerical methods because the continuum theory fails in the nanoscale. Hence, there is a necessity to explain the spreading characteristic of droplets undergoing collisions with solid substrates by an effective method. Many-body dissipative particle dynamics (MDPD) [22] is a widely used mesoscopic scale method with high computing efficiency that can be employed to describe the spreading processes of nanodroplets. Owing to it accurately exhibiting the vapor-liquid coexistence as well as the liquid-solid interaction, MDPD has been used to simulate different droplet behaviors, such as the droplet wetting [23,24], the droplet dynamics [25–27] and the droplet breakup [28].

In the paper, the spreading processes of Newtonian nanodroplets and shear-thinning non-Newtonian nanodroplets with different viscosities on smooth or textured substrates have been investigated through the MDPD method. For the Newtonian nanodroplet, the viscosity is significantly affected by the dissipation parameter of the particles. The maximum spreading diameters of the droplet at different Reynolds and Weber numbers have been analyzed. The polymer droplets have been adopted to reveal the unique spreading characteristic of a shear-thinning non-Newtonian fluid. Additionally, the nanodroplets spreading on textured substrates with diverse pillared structures have been investigated to determine the role of surface roughness.

2. Methods and model configurations

2.1. Many-body dissipative particle dynamics

MDPD is a molecular method developed from the dissipative particle dynamics (DPD) [29]. In DPD, a particle represents clusters of several fluid atoms/molecules, termed as a coarse particle. Compared to DPD, a local density term is added into the conservative force in MDPD to describe the interfacial properties. The interaction forces between two coarse particles consist of the conservative, dissipative, and random forces as represented by F_{ij}^C , F_{ij}^D , and F_{ij}^R , respectively. The specific expressions are given by

$$F_{ij} = \sum_{i \neq j} F_{ij}^C + F_{ij}^D + F_{ij}^R \quad (1)$$

$$F_{ij}^C = A\omega_c(r_{ij})\mathbf{e}_{ij} + B(\rho_i + \rho_j)\omega_d(r_{ij})\mathbf{e}_{ij} \quad (2)$$

$$F_{ij}^D = -\gamma\omega_D(r_{ij})(\mathbf{e}_{ij}\mathbf{v}_{ij})\mathbf{e}_{ij} \quad (3)$$

$$F_{ij}^R = \sigma\omega_R(r_{ij})\xi_{ij}\Delta t^{-1/2}\mathbf{e}_{ij} \quad (4)$$

where $\mathbf{e}_{ij} = (\mathbf{r}_i - \mathbf{r}_j)/|\mathbf{r}_i - \mathbf{r}_j|$ and $\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j$ denote the relative position vector and the relative velocity vector between the particles i and j . All forces work only within a given cutoff radius of r_c . A and B are the attractive and repulsive coefficient, respectively, associated with the surface tensions of the fluids. In Eq. (2), the two weight functions are defined as $\omega_c = 1 - r_{ij}/r_c$ and $\omega_d = 1 - r_{ij}/r_d$ that becomes zero when the specified cutoff radius is exceeded. The first term represents a soft pair potential attractive force and the second term denotes the repulsive interaction. The repulsive force is expressed as a density-dependent

term within an interaction range of r_d . The local density for each particle is given by $\rho_i = \sum_{i \neq j} \omega_p(r_{ij})$ and $\omega_p(r_{ij}) = 15(1 - r_{ij}/r_d)^2/2\pi r_d^3$.

In Eq. (3) and Eq. (4), γ and σ describe the dissipative intensity and the random force, respectively. ω_D and ω_R are r -dependent weight functions that can be defined as $\omega_D(r_{ij}) = \omega_R(r_{ij})^2 = (1 - r_{ij}/r_c)^2$. ξ_{ij} is the Gaussian white noise with zero mean and unit variance, and Δt refers to the timestep of the simulation. The dissipation factor and the random force factor are set to satisfy the fluctuation–dissipation theorem [30] for $\sigma^2 = 2\gamma k_B T$, where k_B is the Boltzmann constant and T is the absolute system temperature determined by the thermostat.

2.2. Non-Newtonian nanodroplet

The finitely extendable nonlinear elastic (FENE) chain was widely adopted to model the real polymer droplet motion that exhibits a shear-thinning property, and could be used to treat non-Newtonian droplets [31–34]. The bead-spring model for polymer chains is proposed by Zhao et al. [35] linking specific MDPD particles together with additional spring forces. In the simulation, polymer chains are randomly distributed among the droplet particles. The additional spring force can be calculated by [36]

$$E = -0.5KR_0^2 \ln \left[1 - \left(\frac{r}{R_0} \right)^2 \right] + 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] + \epsilon \quad (5)$$

The first term in the right represents the attractive force, where K is the spring constant and R_0 denotes the maximum extent of the bond between the particles as a maximum value of the spring extension r . The second term is repulsive within a minimum cutoff of $2^{1/6}\sigma$ of the Lennard-Jones potential. σ and ϵ are the distance and energy, respectively, which are usually set equal to 1.0. The corresponding spring force becomes infinite when r increases to the limited cutoff R_0 that induces the broken-up of molecular bonds.

With the bead-spring chains added into a nanodroplet, polymer chains of the same length are randomly assigned to the droplet particles. The volume fraction of the polymer droplets can be defined as $\phi = N_{\text{bead}} \times N_{\text{chain}} / N_{\text{MDPD}}$, where N_{bead} represents the number of particles on each chain, N_{chain} denotes the number of chains, and N_{MDPD} is the number of the total particles in the system. In order to simulate the transition from the Newtonian droplet to the non-Newtonian droplet, the volume fraction ϕ is set as 5%, 15%, and 30%. The coefficients in Eq. (5) are $K = -30$ and $R_0 = 1.5r_c$. The forces between particles should be recalculated to better consider the effect of polymer chains.

2.3. Modeling parameters

The simulation box is a three-dimensional region consisting of the droplet and the substrate. A cluster of fluid particles with 32,000 particles is filled into the cubic space with a size of $20 \times 20 \times 20$ in MDPD unit, and the vertical distance of the droplet to the substrate, $H = 20$, is given in Fig. 1. The solid substrate, with dimensions $80 \times 80 \times 2$, is perpendicular to the z -axis consisting of 64,000 frozen particles. Under the MDPD interaction, a spherical droplet with the diameter, $D_0 = 24.67$, is spontaneously formed after a relaxation process of 2000 timesteps before the system is equilibrated. The gravity is usually ignored at the nanoscale. The process of droplet spreading is initialized through assigning a vertical velocity V to the group of droplet particles, in which the droplet particles interact with the substrate particles to reach a finally stable state.

Fig. 2 shows a specific display of the shape of the textured substrates decorated by pillared structures, which are differed by the surface fraction ($f = l^2/L^2$) and the roughness factor ($r = 1 + 4lh/L^2$) [37]. The size of the relative square substrate is fixed with $L = 3$ and the height of each column is set as $h = 1.5$. By changing the width of $l = 1.0, 1.5, 2.0, 2.5$, the surface fraction is calculated as $f = 0.11, 0.25, 0.44, 0.69$ and the roughness factor $r = 1.67, 2.00, 2.33, 2.67$. The two finally

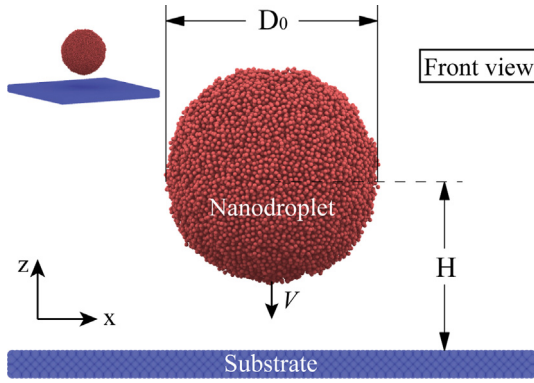


Fig. 1. A typical initial simulation state describing the nanodroplet spreading process.

stable states, including the Wenzel state and the Cassie-Baxter state, are observed, and work as an important evaluation for identifying the spreading behavior of the droplets on the textured substrates.

In MDPD simulation, the interaction function and the corresponding parameters describing the pairwise interactions between particles are particularly crucial for accurate prediction of the fluid properties, mainly related to the attractive coefficient A and the repulsive coefficient B . A_{ll} and B_{ll} indicate the interaction between different liquid (l) particles, which determines the inherent surface tension of a droplet, while A_{sl} and B_{sl} describe the interaction between solid (s) particles and liquid (l) particles associated with the wettability of a substrate. The surface tension of a droplet can be obtained by the fitting equation modified by Zhao et al. [35],

$$\sigma^{fit} = -\frac{\pi}{240}(0.376Ar_c^5\rho^2 + 0.003Br_d^5\rho^3) \quad (6)$$

The coefficients A and B specified in Eq. (6) are strongly related to the surface tension. Besides, the kinematic viscosity of the fluid is mainly derived from the diffusion motion of particles and the viscous dissipation energy within particles as [29]

$$\nu = \frac{\eta_{diff} + \eta_{diss}}{\rho} = \frac{45k_B T}{4\pi\gamma\rho r_c^3} + \frac{2\pi\gamma\rho r_c^5}{1575} \quad (7)$$

Eq. (7) shows the relationship between the viscosity and the simulation parameters. The system temperature $k_B T$ is utilized to be a thermostat relating to the system thermodynamic equilibrium in MDPD simulation, which could not be modified adjusting the viscosity of the droplet. Additionally, the required calculation time is proportional to

Table 1

Parameters used in MDPD simulations.

Parameter	Symbol	Value
Attractive coefficient (ll)	A_{ll}	-40
Attractive coefficient (sl)	A_{sl}	-25
Repulsive coefficient	$B_{ll} = B_{sl}$	25
Particle density	ρ	6.0
System temperature	$k_B T$	1.0
Dissipation parameter	γ	18.0
Random parameter	$\sigma^2 = 2\gamma k_B T$	6.0
Cutoff radius	r_c	1.00
Repulsive interaction range	r_d	0.75

the cutoff radius. Hence, the dissipation parameter γ is finally used in controlling the viscosity of the droplet. During simulations, A_{sl} and B_{sl} are set for a moderate wetting characteristic of the substrate with a static contacting angle of about 86° . All parameters are listed in Table 1.

Dimensionless units are usually used in MDPD simulations to avoid computational deviation during data processing. The parameters listed in Table 1 are all given in MDPD units, and the real units need to be recalculated. Owing to that one MDPD particle represents a cluster of fluid molecules, the mass of an MDPD particle is regarded as the fundamental unit of mass. To satisfy the transport properties of the fluid, especially the viscosity and the surface tension, actual units should be particularly considered. The choice of the value $\gamma = 10$ is corresponding to the kinematic viscosity of the water, and the unit can be described as $\nu^* = 3.364 \times 10^{-6} \text{ m}^2/\text{s}$. To satisfy the unit of the surface tension, the other properties of droplets are matched by $M^* = 2.71 \times 10^{-24} \text{ kg}$ (Unit mass), $L^* = 8.6 \times 10^{-10} \text{ m}$ (Unit length), and $T^* = 2.2 \times 10^{-11} \text{ s}$ (Unit time), where the conversion from MDPD units to the physical units of the parameters can be obtained. All simulation details are listed in Table 2.

During the simulations, the positions and velocities of particles are updated using the modified velocity-Verlet time-integration algorithm with a constant time step of $\Delta t = 0.01$ and a relaxation parameter of $\lambda = 0.65$ [38]. All simulations are performed by the widely used molecular dynamics package LAMMPS [39], and simulation results are exhibited by Open Visualization Tool (OVITO) [40]. The substrate atoms are frozen to represent the boundary wall, which prevents particles from escaping and improves the computational efficiency significantly. Each case is typically equilibrated for 2×10^3 timesteps before running for at least 2×10^4 timesteps.

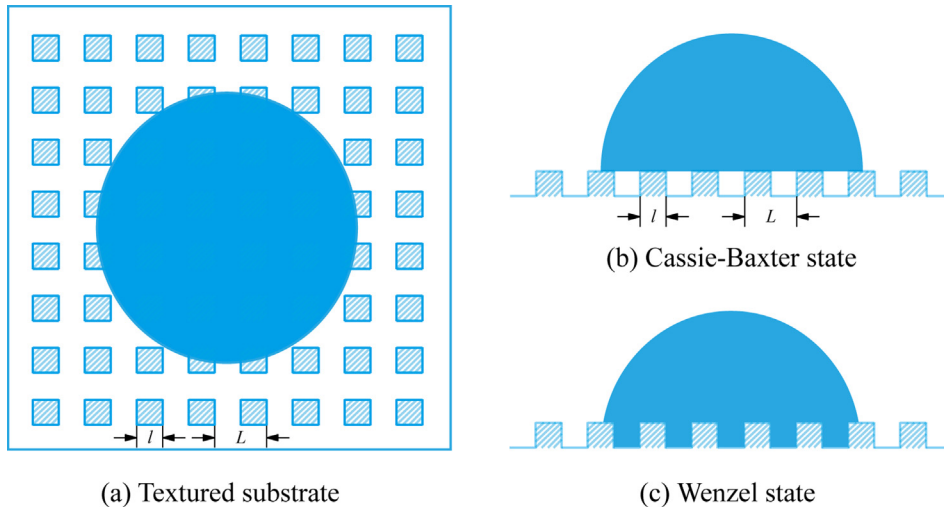


Fig. 2. The droplet states on textured substrates. (a) The textured substrate defined by the surface fraction; (b) the Cassie-Baxter state; (c) the Wenzel state.

Table 2
Simulation details in the present study.

Set	Fluid type	Droplet diameter (10 ⁻⁸) (m)	Dissipation parameter	Kinematic viscosity (10 ⁻⁶) (m ² /s)	Surface tension (10 ⁻³) (N/m)	Substrate type	Static contact angle	Volume fraction	Surface fraction
1	Newtonian fluid	2.12	$\gamma = 1, 5, 10, 18, 36, 60$	$\nu = 2.09, 0.80, 1.00, 1.56, 2.95, 4.86$	39.41	Flat	86°	–	1
2	Non-Newtonian fluid	2.12	$\gamma = 18$	$\nu = 1.56$	39.41	Flat	86°	5% 15% 30%	1
3	Newtonian fluid	2.12	$\gamma = 5, 18, 36, 60$	$\nu = 0.80, 1.56, 2.95, 4.86$	39.41	Textured	55° 115°	–	0.11 0.25 0.44 0.69

3. Results and discussion

Nanodroplet spreads by giving a vertically downward acceleration to arrive at the substrate. Then, the droplet with the shape of a thin cylindrical disk is rapidly formed within several milliseconds under the inertial and capillary forces. These forces are usually evaluated by dimensionless numbers, i.e., the Weber number, $We = \rho D_0 V^2 / \sigma$ and the Reynolds number, $Re = VD_0 / \nu$, where ρ is the density of fluid particles, D_0 is the initial diameter of droplets, and V is the droplet impacting velocity.

The dynamic spreading and receding processes of a nanodroplet are illustrated in Fig. 3. Two initial velocities are given to compare the difference of the spreading processes under variable Weber numbers. When a droplet impinges the substrate, the spreading process has arisen rapidly to reach the maximum contact diameter. Fig. 3 shows that the maximum spreading diameter is generally larger under a higher Weber number, and the relevant time taken for reaching the maximum diameter is less than the case under a lower Weber number, which tells that the inertial force is more dominant than the viscous dissipation and surface tension during the early stage of the spreading. Additionally, the final states of the nanodroplets are found independent of the initial kinetic energy as depicted in Fig. 3.

3.1. Newtonian nanodroplet spreading with different viscosities

3.1.1. Influence of viscosity

The function of the dissipation coefficient γ is reflected in Eq. (7). Based on the expression for the calculation of viscosity, the increase of the dissipation parameter γ improves the second term on the right of the equation (the dissipation term), and enhancement of the system temperature $k_B T$ increases the first term (the diffusion term). The dimensionless spreading parameters $\beta = D/D_0$ and $\xi = H/H_0$ are used to assess the quality of the spreading process, where D_0 and H_0 denote the initial diameter and height of the droplet, respectively. When giving an impacting velocity $v = 1$, the dynamic spreading diameter and the droplet height have been calculated by varying the dissipation

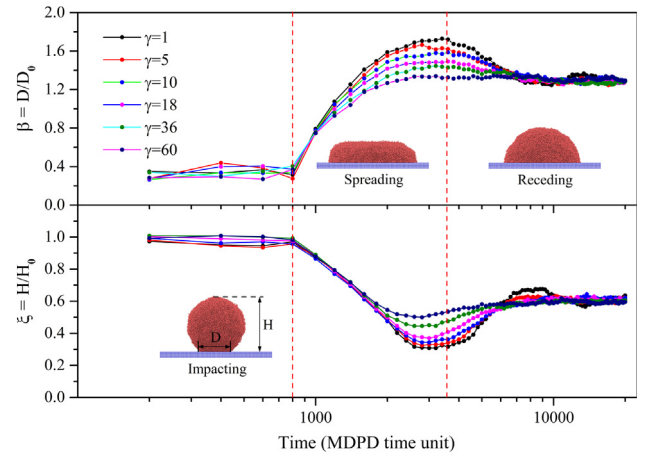


Fig. 4. The time-dependent spreading parameters of droplets with different viscosities.

coefficient of $\gamma = 1, 5, 10, 18, 36$ and 60 (see Fig. 4). The moving process of the droplet could be described by three major stages, that is, the impacting stage, the spreading stage, and the receding stage. The shape of nanodroplets remains nearly spherical in the early stage before it impacts the substrate. A strong deformation occurs as the pressure increases rapidly, which enables a sharply instantaneous change of the spreading parameters. The nanodroplets finally arrive at a stable state. As γ increases, the maximum spreading diameter decreases due to the influence of viscosity. Under higher viscosity, the curve is smoother with a lower fluctuation demonstrated by a clear variance in the receding regime. This phenomenon indicates that the influence of viscous dissipation becomes significant than other internal forces during the receding stage. Moreover, the time for more viscous droplets to reach a stable state is much shorter.

The spreading curve in Fig. 4 is compared with the values calculated theoretically by Eq. (7). There rises a special point at $\gamma = 1$ inconsistent

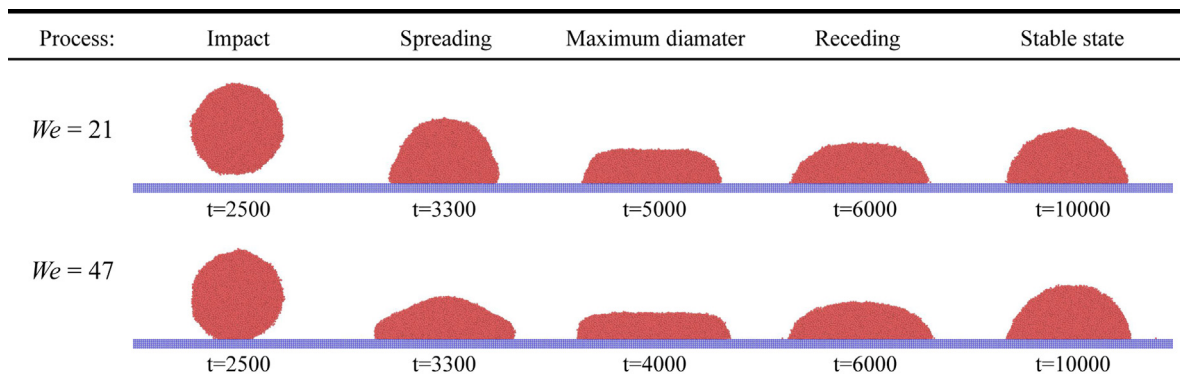


Fig. 3. Snapshots of inertial droplets spreading on the flat surfaces for $We = 21$ and $We = 47$.

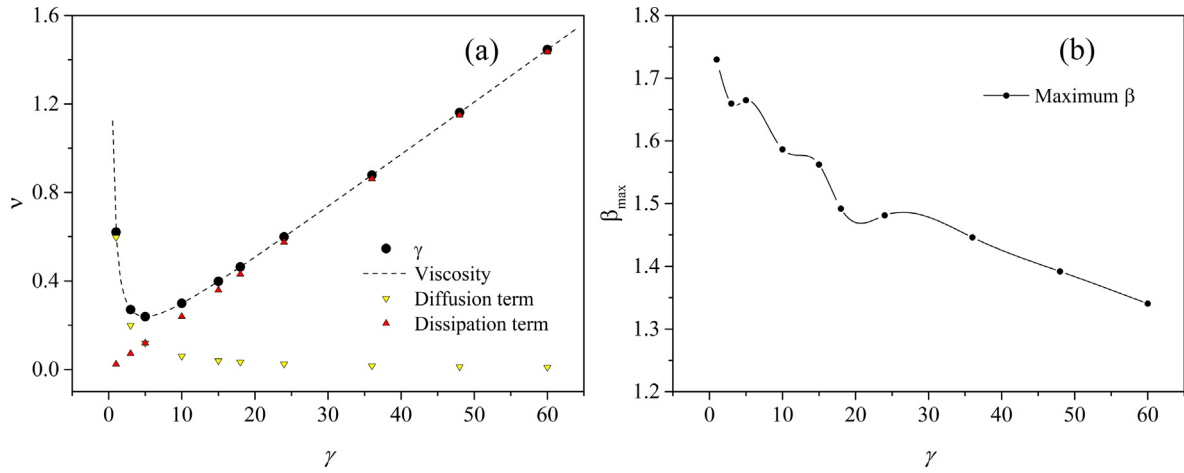


Fig. 5. (a) The theoretical viscosities of droplets. (b) The corresponding maximum spreading diameters of droplets.

with the theory. Fig. 5 shows the theoretical viscosity and the corresponding maximum diameter by changing the dissipative coefficient γ . The diffusion and dissipation terms play disparate roles in constituting the viscosity of the droplets. When γ is less than 5, the diffusion motion of fluid particles is the major factor affecting the viscosity. A higher viscous droplet with $\gamma = 1$ ought to obtain a relatively smaller spreading diameter, which is absolutely contrary to the actual result. The trend of the maximum spreading diameters is more related to the diffusion term. The special phenomenon can be explained by the strenuous motion of the molecule gas-like behavior. Boromand et al. [41] proposed that the Groot–Warren scaling law for deciding viscosity leads to erroneous results at low dissipative parameters. When the diffusion term is a dominant factor in the driving force, the droplet particles tend to be more dispersed mainly responsible for the higher spreading diameter.

3.1.2. Maximum spreading of a Newtonian droplet

In order to reveal the relationship between the maximum spreading diameters and the initial impacting parameters, the dissipation parameter γ is set as 5, 10, 18, 36 and 60 to provide different viscous droplets. Other parameters are fixed except for the velocity that is associated with the Weber number and the Reynolds number. The impacting velocity is set as $v = 1.0, 1.5, 2.59, 3.66$ and 5 regarding $We = 21, 47, 141, 282$ and 526 . Fig. 6(a) shows a prediction method of the maximum spreading diameters of nanodroplets undergoing

collisions with moderate solid surfaces at low Reynolds numbers. The physical terms are combined to give an expression of the maximum spreading ratio $\beta_{\max}$ as

$$\beta_{\max} = 0.7Re^{0.1445}We^{0.066} \quad (8)$$

where the Reynolds number ranges from 17 to 378, and the Weber number ranges from 21 to 526. The expression gives a straight line. The experimental data obtained by An and Lee [42] for water and glycerin impacting a smooth stainless steel are used to test the accuracy of the expression, and the existing maximum spreading models for Newtonian droplets on the order of millimeter are used to compare the dynamics between the nanodroplets and the bigger droplets as listed in Table 3. Scheller and Bousfield [43] model is based on the regression analysis of the experimental data, and Roisman [44] model is an analytical self-similar solution for the viscous flow. In their models, the spread is generated by normal droplet impacting onto a dry substrate with high Weber and Reynolds numbers. Fig. 6(b) shows the comparison between the experimental data and the predicted maximum spreading diameters by the models. The prediction of our simulation is found to give a better fit at low maximum spreading conditions, and the predicted maximum spreading is found to be smaller than experimental data with high Reynolds numbers. The underprediction of our model at high Reynolds numbers is explained by the dominant effect of the viscosity in the nanoscale, which is different from the bigger droplets. The predicted equation is found to fit quite well under the current ranges of the

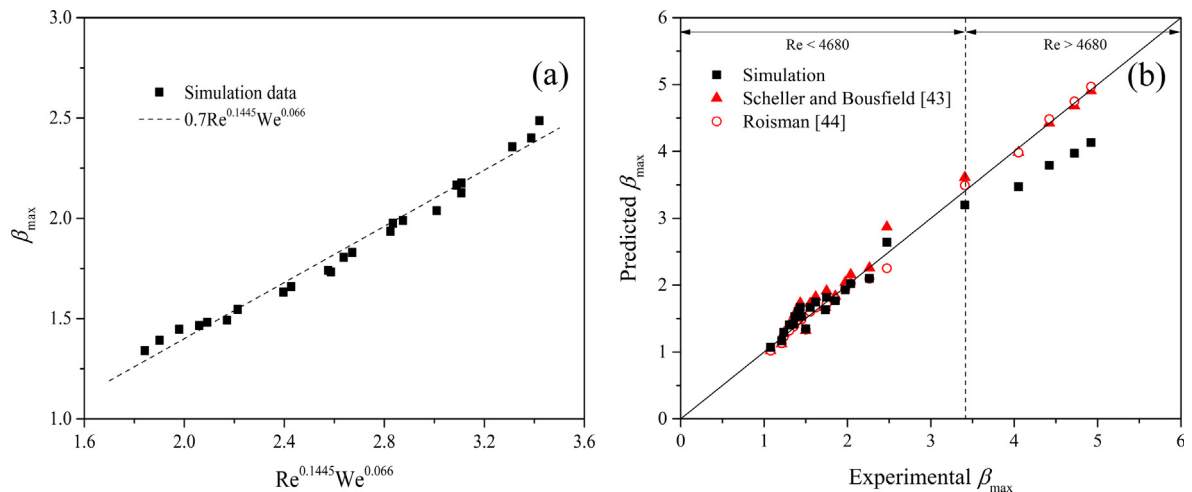


Fig. 6. (a) The fitting curve between the maximum spreading diameter and the dimensionless number. (b) Comparison between the experimental data and the predicted results.

Table 3
Maximum spreading models and experimental data for Newtonian droplets.

Researchers	Predictions	D_0 (m)	Fluid	Re	We
An and Lee [42](Experiment)	–	2.5×10^{-3}	Water and glycerin	4–11786	23–791
Our simulation	$\beta_{\max} = 0.7Re^{0.1445}We^{0.066}$	2.12×10^{-8}	–	17–378	21–526
Scheller and Bousfield [43]	$\beta_{\max} = 0.61(Re^2Oh)^{0.166}$	3.0×10^{-3}	–	19–16400	90–1973
Roisman [44]	$\beta_{\max} = 0.87Re^{1/5} - 0.4Re^{2/5}We^{-1/2}$	1.0×10^{-3}	–	12–36758	4–5722

dimensionless numbers, and can be applied in droplet spreading with high viscosities and small Reynolds numbers.

3.2. Spreading of a non-Newtonian droplet

Apparently, as mentioned in Section 3.1, the viscosity greatly influences the spreading process of droplets. The previous study is typically concentrated on a Newtonian droplet, while the viscosity of a non-Newtonian droplet tends to be more unpredicted owing to the complex rheology of the fluid. In order to couple the spreading characteristic with the mesoscopic simulation, the finitely extendable nonlinear elastic (FENE) chain was added into the original droplet particles to create a continuum model of the polymeric droplet that exhibits a shear-thinning non-Newtonian property. The interaction between the polymer chains is recalculated for an efficient simulation.

The existing studies about the non-Newtonian properties of polymer droplets are usually verified by the shear flow [31–33 45]. Reverse Poiseuille flow (RPF) is a well-known case to identify the viscous flow of complex fluid. In the present study, the cubic space with a size of $20 \times 20 \times 20$ containing 32,000 particles is chosen to simulate the shear flow. The modeling region is divided into two equally upper and lower parts. The two parts are sheared by two forces with the same level in the opposite direction to create an identical shear effect. The volume fraction ϕ is set as 5%, 15%, and 30%. The fully-developed velocity profiles along the z-direction for three types of non-Newtonian fluids and a Newtonian fluid are shown in Fig. 7(a). The non-Newtonian fluid has a lower velocity than the Newtonian fluid, which denotes that large viscous force inhibits the motion of the fluid particles. The corresponding result can be described by the power-law of the Poiseuille flow as

$$V(z) = V_{\max} \left[1 - \left(\frac{z}{h} \right)^{1+\frac{1}{n}} \right] \quad (9)$$

$$V_{\max} = \frac{n}{1+n} \left(\frac{\rho f}{\kappa} \right)^{\frac{1}{n}} h^{1+\frac{1}{n}} \quad (10)$$

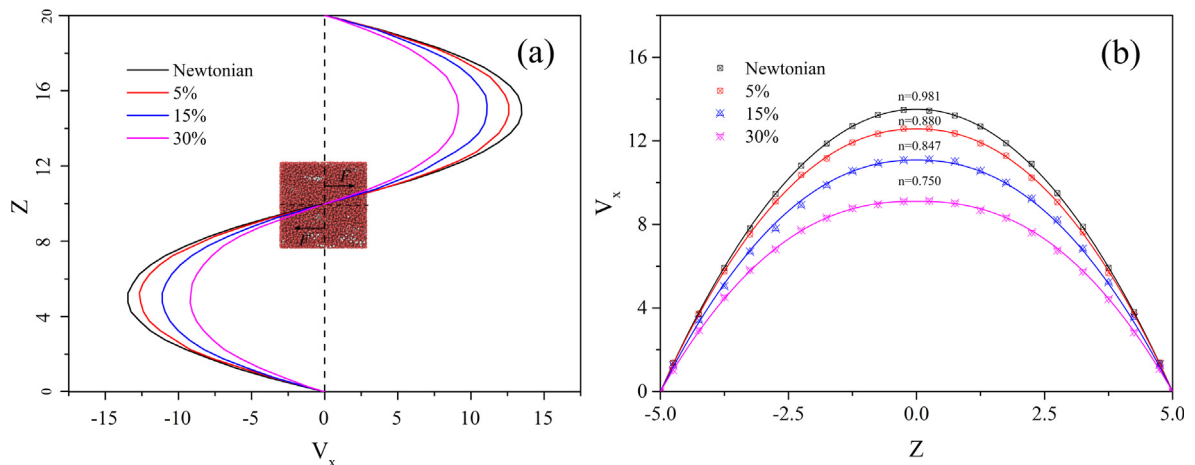


Fig. 7. (a) The fully-developed velocity profiles of the Newtonian and the non-Newtonian fluids with the volume fractions $\phi = 5\%$, 15% , and 30% . (b) The power-law indices calculated by curve fitting according to the velocity profiles $n = 0.981, 0.880, 0.847, 0.750$, correspondingly.

where n is the power-law index, ρ denotes the particle density, f indicates the shear force, κ is the power-law shear stress coefficient, and h represents the half-width of the region. When $n = 1$, Eq. (9) can be simplified to a common equation of the Newtonian fluid. When n less than 1, the equation is used to describe a shear-thinning fluid. The fitting values of the power-law index n regarding the curves in Fig. 7(a) are drawn in Fig. 7(b). The volume fraction $\phi = 5\%$, 15% and 30% correspond to $n = 0.880, 0.847$ and 0.750 . Conclusively, the droplets fluid with polymer chains certainly exhibits the non-Newtonian behavior.

To identify the viscosity of the polymer droplets, the apparent viscosity is adopted for the non-Newtonian fluid. Fedosov et al. [31] proposed that for the low shear rates near the centerline, the velocity profiles are fitted well with

$$V(z) = V_{\max} - \frac{\rho f}{2\eta_0} z^2 + O(z^4) \quad (11)$$

where η_0 is the shear viscosity. The shear forces $f = 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7$ and 1.0 , are imposed to investigate the influence of viscosity as shown in Fig. 8(a). For the Newtonian droplets, there are virtually no changes in dynamic viscosity. For the polymer droplets, however, the shear-thinning property is predicted to enhance the fluid viscosity as the volume fraction of the polymer increases. The obvious decrease of viscosity is due to the increase of the shear force, which enlarges the polymer chains and adjusts the spring force. The polymer chains usually curl up into irregular clusters. When giving an external force, the polymer chains stretch out along the direction of the force, which releases the elastic potential energy of the chains. As the shear force increases, the curve approaches at a slow rate. Therefore, the viscosity of the droplets with polymer chains is obviously much larger than the common Newtonian droplet, which reducing by the increase of the force.

The spreading processes of the nanodroplets as mentioned above are simulated, and the relevant maximum spreading diameters are presented in Fig. 8(b). For the given impacting velocity $v = 0.53, 1, 1.5, 2.59, 3.66, 5$ and 6 , regarding the Weber number $We = 6, 21, 47, 141$,

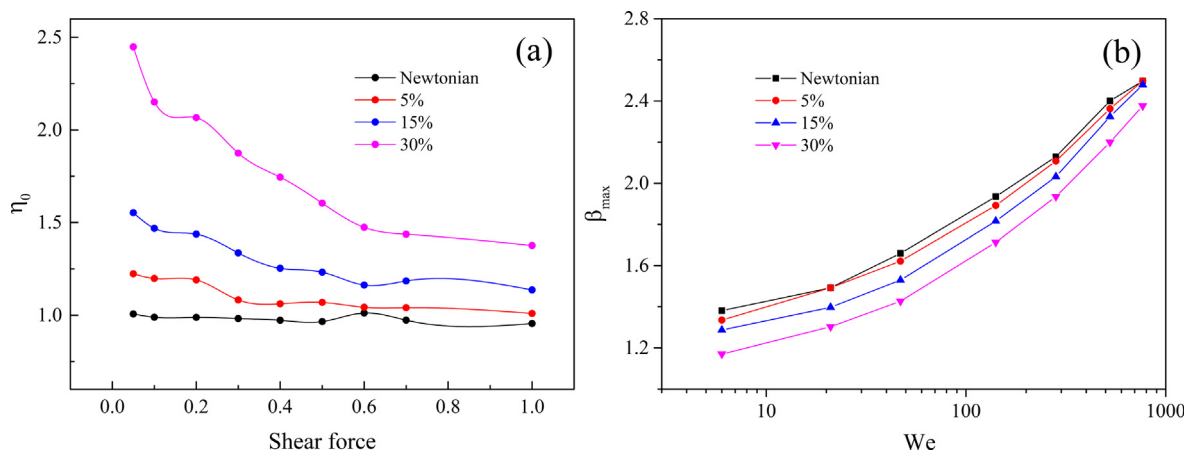


Fig. 8. (a) Influence of the viscosity for different droplets by increasing the shear forces. (b) The maximum spreading diameters of the Newtonian droplets and the non-Newtonian droplets with different volume fractions of the polymer.

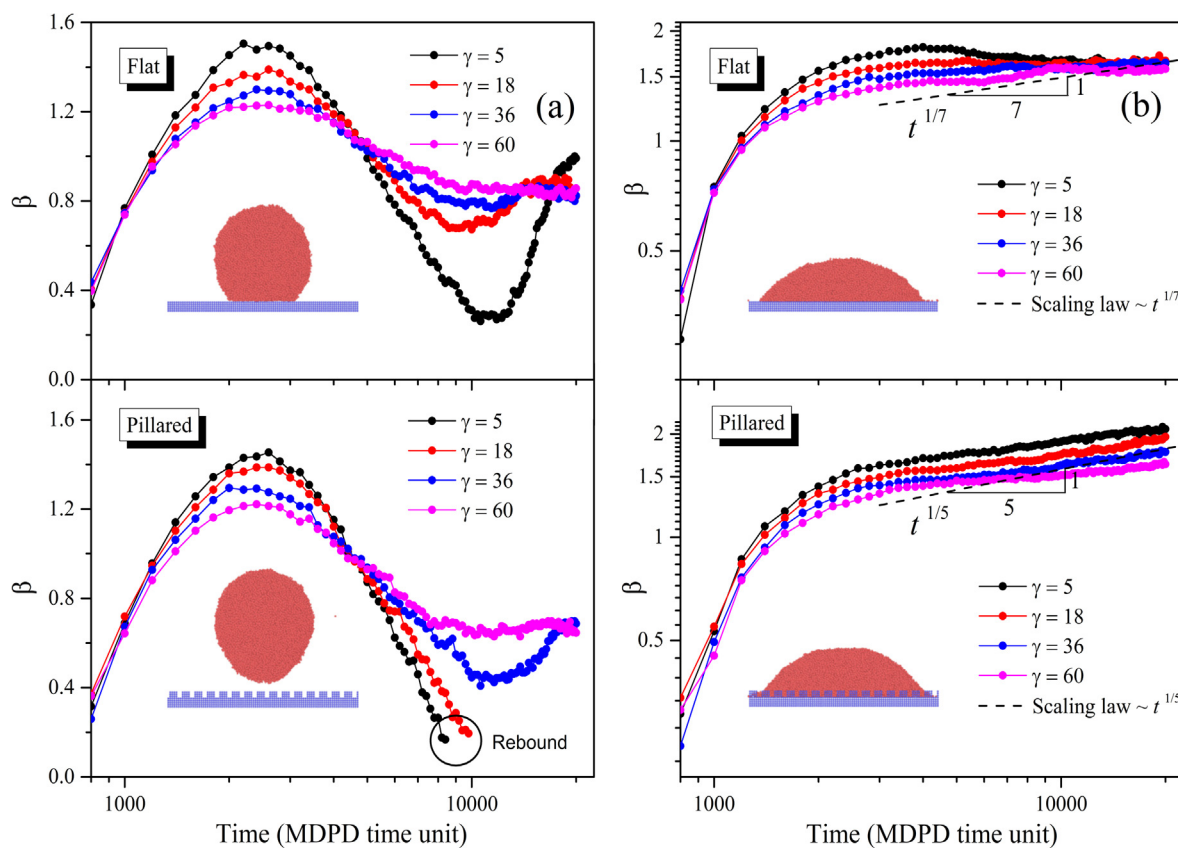


Fig. 9. The dynamic spreading diameters of droplets with different viscosities on the flat substrates and the pillared substrates for (a) the hydrophobic substrate and (b) the hydrophilic substrate. The dashed black lines represent the scaling laws $\beta \sim t^{1/7}$ and $\beta \sim t^{1/5}$, respectively.

282, 526 and 757, the maximum spreading diameters of the droplets have been predicted. The larger the volume fraction is, the smaller the maximum spreading diameters are. As the shear effect occurs under the high inertial force, $\beta_{\max}$ seems gradually consistent with each other except for the case with $\phi = 30\%$. For a much larger volume fraction, the droplet spreading process is strongly suppressed. The polymer chains need more energy to release, which leads to a decrease of viscosity.

3.3. Spreading of a nanodroplet on a textured substrate

The surface roughness is further considered owing to the anisotropy

of patterned substrates. Here we focus on the effect of the surface roughness decorated by different pillared structures. Since the structured substrates are widely applied to create the superhydrophilic or superhydrophobic surfaces, the hydrophilic and hydrophobic substrates (a static contact angle is about 55° and 115°) are chosen to examine effects of the pillared structures. Moreover, the influence of the viscosity is also especially considered.

With the given initial impacting velocity $v = 1$ and the specific surface fraction $f = 0.44$, the comparison of the droplets spreading on the flat substrates and on the pillared substrates is shown in Fig. 9. In the spreading stage dominated by the inertial force, there is no significant difference in the predicted curves. However, the receding stage

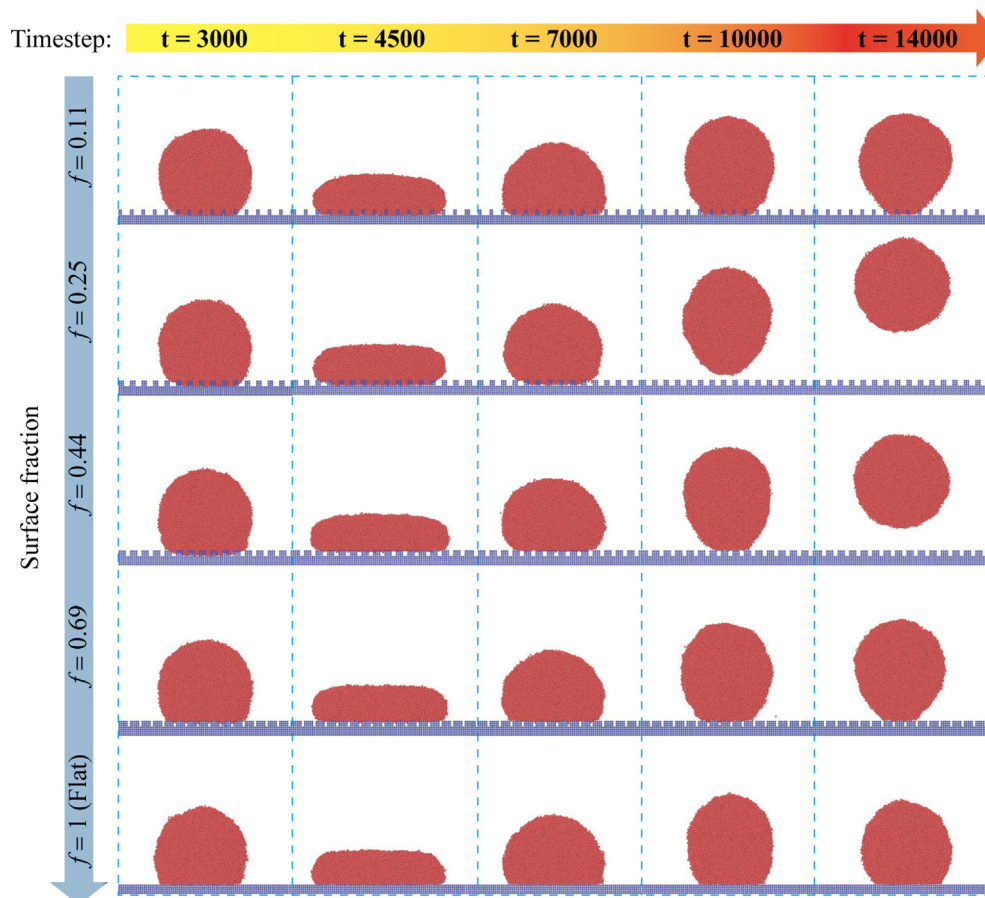


Fig. 10. Snapshots of the droplet shapes spreading on the textured substrates with increasing surface fractions.

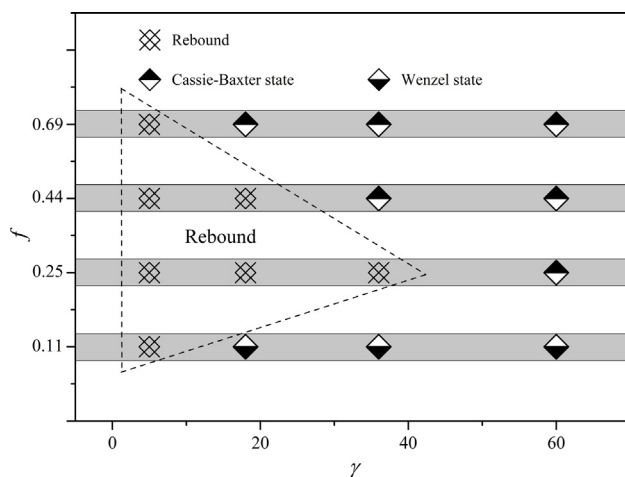


Fig. 11. The final states of the droplets with different viscosities on the textured substrates which shows the region of rebounding behavior.

is extremely altered by the textured surfaces, contrary to that of the flat surfaces. As shown in Fig. 9(a), with the lower viscosity, the droplets rebound on the hydrophobic pillared substrates due to the commonly initial kinetic energy, while the droplets keep stable on the flat substrates. The pillared structures profoundly raise the intrinsic wetting characteristics of the substrates, which makes the substrate more hydrophobic. As the viscosity increases, the expansion effect obviously weakens. The phenomenon is also reflected in Fig. 9(b) for the droplets spreading on hydrophilic substrates. The droplets easily spread over the substrates, and keep moving further to occupy the entire pillared holes,

accompanied by the Wenzel state. Meanwhile, the viscous force prevents the droplets from continuous spreading. Besides, as shown in Fig. 9(b), with different fluid viscosities, the spreading diameters approximately obey the scaling law of $\beta \sim t^{1/7}$ for the hydrophilic flat substrates in the wicking process, and obey the scaling law of $\beta \sim t^{1/5}$ for the hydrophilic pillared substrates, which are in agreement with the previous study of Yuan and Zhao [46]. They obtained the scaling law by both the experiments and the simulations, and pointed out that the scaling exponent would greatly change from about $1/7$ to about $1/3$ when the smooth substrate changed to a pillar-arrayed substrate. The scaling exponent depends on the ability of pillared substrates to enhance the surface wettability. Therefore, the pillared structures approve to highly enhance the intrinsic wetting characteristics of the substrates regardless of the surface wettability, which can be controlled by increasing the viscosity of droplets.

Besides the pillared structure, the surface fraction f is also taken into consideration. Fig. 10 shows the droplet spreading processes on the substrates with different surface fractions. The surface fraction f is selected as 0.11, 0.25, 0.44 and 0.69. The dissipation parameter is fixed to $\gamma = 18$, that is, with a constant fluid viscosity, and the velocity is fixed to $v = 1$. For $f = 0.25$ or 0.44 , the droplets rebound on the substrates, which is in agreement with the experimental study of Patil et al [14]. The rebounding stage occurs more quickly under $f = 0.25$. The surface fraction is tightly associated with the droplet behavior, rebound or keeping stable on the substrates. As the surface fraction approaches the value of 0.25, the expansion effect is the most dominant factor. With a smaller surface fraction, the final state of the droplet tends to be the Wenzel state. The Cassie-Baxter state is observed with a higher surface fraction. Moreover, the partial Wenzel state can be caught during the spreading stage. As shown in Fig. 10, when the surface fraction $f = 0.25$ or 0.44 , the Wenzel to Cassie wetting transition is observed. It may be

due to the variance of the contact areas that are defined by the pillared structures, where the surface fraction changes the internal forces of the particles. Additionally, the dynamic contact angle is conspicuously influenced by the pillared structures, which induces the spreading process to be unsteady. Fig. 11 shows the results of different viscous droplets undergoing collisions with four types of substrates. The dissipation parameter is set as $\gamma = 5, 18, 36$ and 60 , which indicates an increasing viscosity. The figure explicitly illustrates the relevance between the fluid viscosity and the pillared structures. When f is 0.25 , the droplet easily rebounds on the substrates except for $\gamma = 60$. This special phenomenon could be well explained by Yuan et al [47]. They proposed the competition between the viscous resistance and molecular friction. When the viscosity is small, the molecular friction dominates. With the increase of the viscosity, the effect of the viscous resistance increases. As the surface fraction increases or decreases further, the rebounding behavior gradually disappears. In general, the wettability expansion effect of the pillared structure shows the most intensive extent for the surface fraction $f = 0.25$, and the high viscosity is found to profoundly weaken this effect.

4. Conclusions

In the paper, the processes of nanodroplets with different viscosities spreading on varied surfaces are simulated by many-body dissipative particle dynamics method. A large viscosity effectively suppresses the spreading actions of the droplets, and makes the droplets to be more stable upon the solid surfaces. The spreading characteristic of nanodroplets is eventually evaluated by the dimensionless maximum spreading diameters associated with the Weber number and the Reynolds number with an equation of $\beta_{\max} = 0.7Re^{0.1445}We^{0.066}$, where the Reynolds number ranges from 17 to 378 and the Weber number ranges from 21 to 526 . The droplets with the polymer chains are analyzed to exhibit a shear-thinning non-Newtonian behavior. The droplets with different volume fractions of the polymer are adopted during spreading on the solid substrates. The corresponding maximum spreading diameters are quite different under a small initial kinetic energy, and gradually become consistent at higher Weber numbers due to a shear-thinning effect. Finally, the influence of the surface roughness differed by the specific pillared structures has been investigated. It is found that the pillared structures profoundly enhance the intrinsic wetting characteristic regardless of the surface wettability. In particular, the wettability expansion effect is the most significant for the surface fraction $f = 0.25$, which is weakened by a large viscosity. The study gives a comprehensive understanding of the nanodroplet spreading characteristic, and reveals the significant influence of the viscosity on the droplet spreading upon intricate surfaces, which provides a useful direction for controlling the droplet behavior.

CRedit authorship contribution statement

Hao Chen: Investigation, Software, Validation, Data curation, Writing - original draft. **Qichun Nie:** Conceptualization, Methodology, Visualization, Resources. **Haisheng Fang:** Supervision, Writing - review & editing, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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